

Junior Makers · Introduction

# Welcome to Robotics!

What robots are, how they work, and your first build: an Art Bot

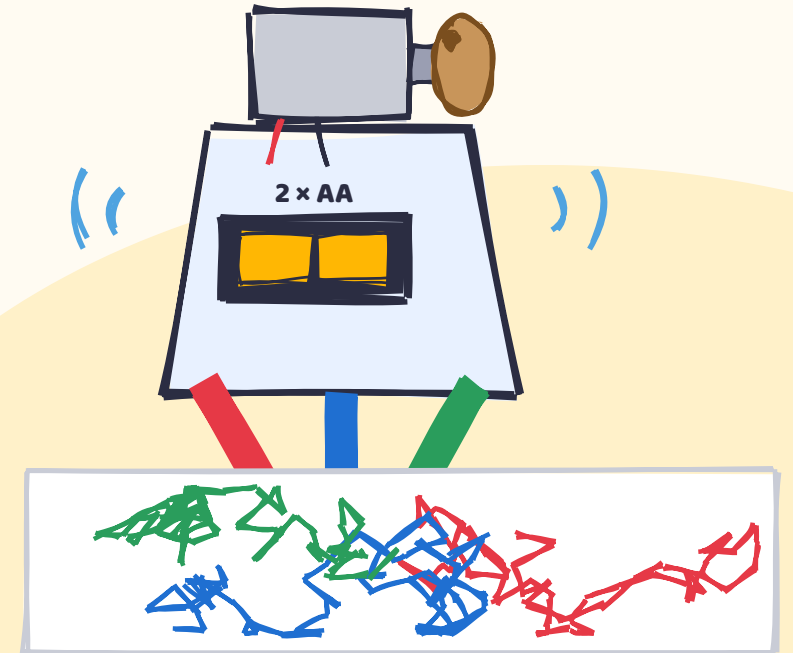
Sense, think, act

Parts of a robot

DC motor + vibration

Did you know?

Your phone vibrates using a tiny motor with an off-centre weight. Today you use the same trick to make art!



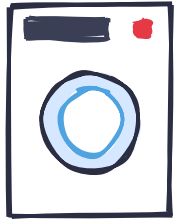


## Quick quiz

# Is it a robot? Thumbs up or down!

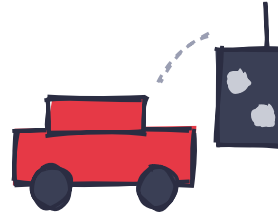


1



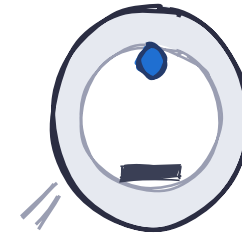
Washing machine

2



Remote-control car

3



Robot vacuum

4



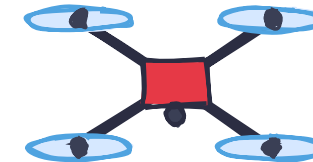
Calculator

5



Self-driving car

6



Delivery drone



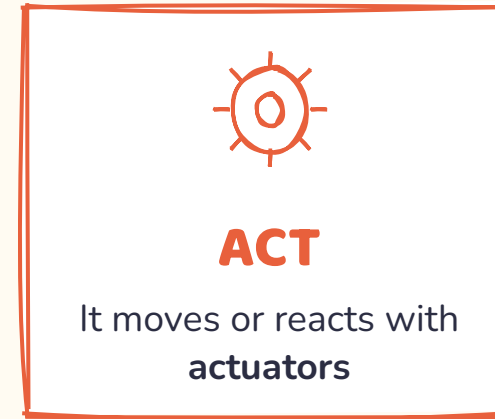
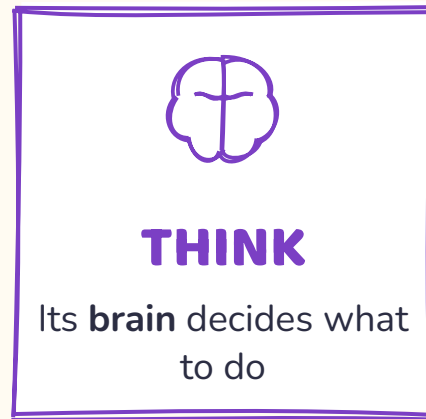
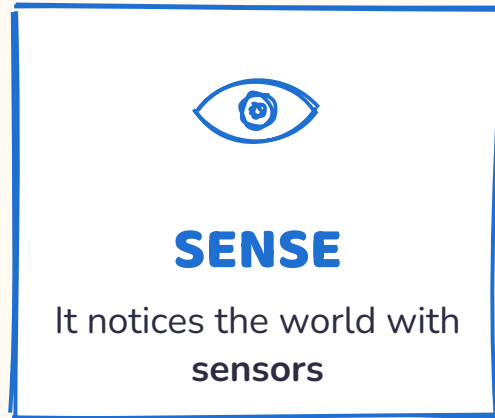
Talk with your partner: **what makes something a robot** and not just a machine?





The big idea

# A robot can SENSE, THINK and ACT



## Our definition

A robot is a **machine** that can **sense** its surroundings, **decide** what to do using a **program**, and then **act**, all **by itself**.

You do it too!

You **SEE** a ball coming, your brain **THINKS** 'catch!', your hands **ACT**.





## Answers



# So... which ones are robots?

**Sort of****Washing machine**

It follows a program and senses water level, but it cannot move around or change plans much.

**No****Remote-control car**

A person does all the thinking. No sensing, no deciding by itself.

**Yes!****Robot vacuum**

Senses walls and dirt, decides where to go, moves by itself.

**No****Calculator**

It thinks (calculates) but cannot sense the world or move.

**Yes!****Self-driving car**

Cameras and radar sense, a computer decides, motors steer and brake.

**Yes!****Delivery drone**

Uses GPS and sensors to fly a route and land on its own.



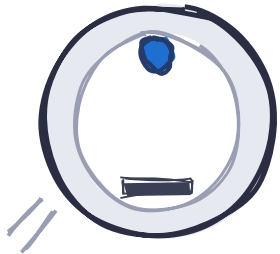
The more a machine can **sense, decide and act on its own**, the more "robot" it is. Engineers call this being **autonomous**.





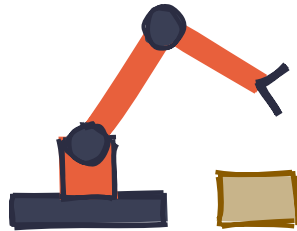
Look around you

# Robots are everywhere!



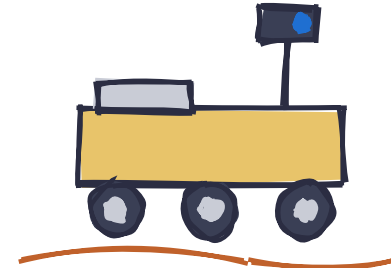
## Home helper

Cleans floors by itself



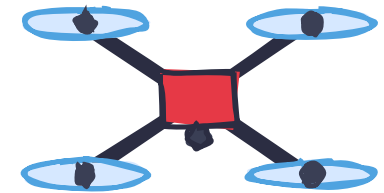
## Factory arm

Builds cars, packs boxes



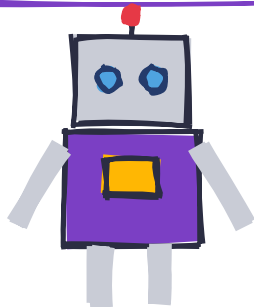
## Mars rover

Explores another planet



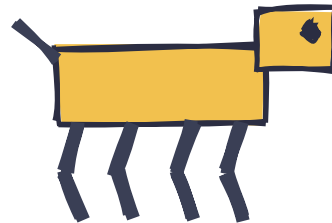
## Drone

Films, maps, delivers



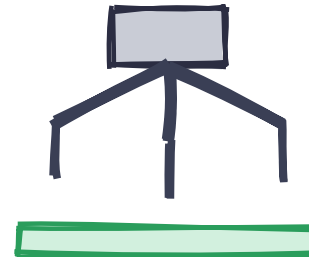
## Humanoid

Walks and talks like us



## Robot dog

Inspects dangerous places



## Surgical robot

Helps doctors operate



## Self-driving car

Drives itself on roads



## A short history

# From an idea to everywhere

**1921**

The word **robot** first appears in a Czech play by Karel Čapek. It comes from *robota*, meaning hard work.

**1997**

**Sojourner**, a rover the size of a microwave, drives on Mars.

**2021**

**Ingenuity** makes the first powered flight on another planet, Mars.

**1961**

**Unimate**, the first factory robot arm, starts work in a car factory in the USA.

**2002**

**Roomba** robot vacuums go on sale and robots enter our homes.

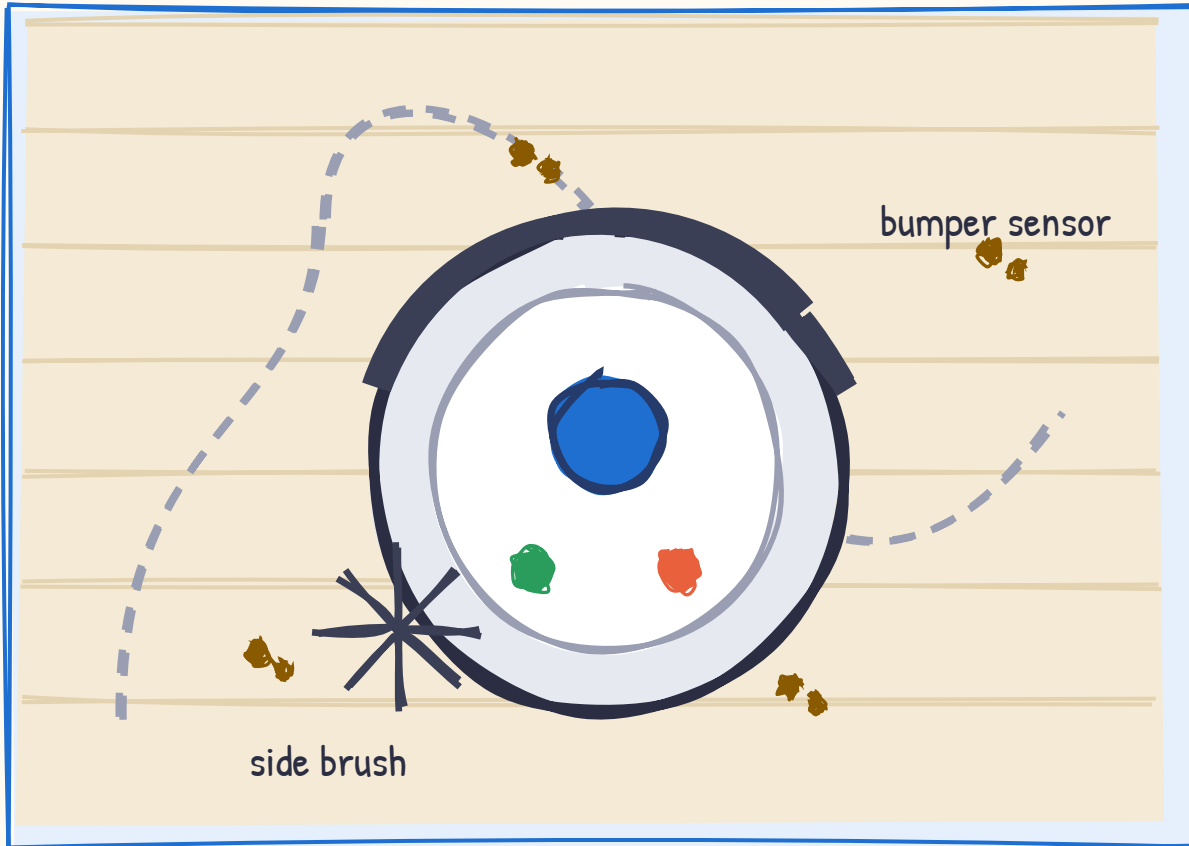
**Today**

Robots work in hospitals, farms, warehouses and space, often with **AI** inside.



Robot of the home · iRobot, USA · 2002

# Roomba



## What it does

A round robot that **vacuums your floor by itself**. It drives around furniture, cleans under beds and goes back to its **charging dock** when its battery is low.

**SENSE:** bump sensors, cliff sensors (stop falls down stairs), dirt sensors

**THINK:** plans a cleaning path

**ACT:** wheels, spinning brushes and a vacuum

## Fascinating fact

iRobot has sold tens of millions of robot vacuums. Many homes have a robot before they have any other robot!



Robot dog · Boston Dynamics, USA

# Spot



## What it does

A four-legged robot that **walks, climbs stairs and opens doors**. It inspects places that are dangerous for people, like power stations, oil rigs and building sites.

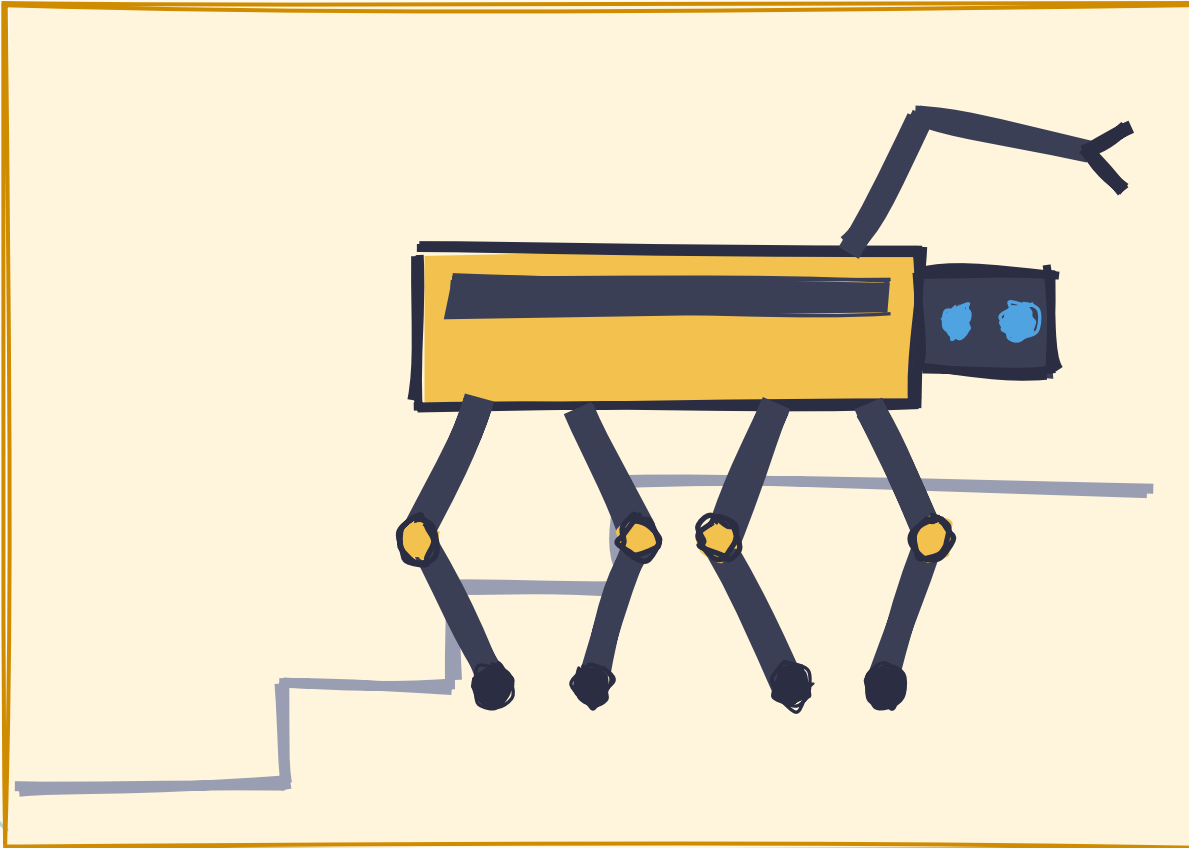
**SENSE:** cameras on every side see in 3D all around it

**THINK:** keeps its balance hundreds of times every second

**ACT:** 12 motors move its 4 legs; an optional arm can grab things

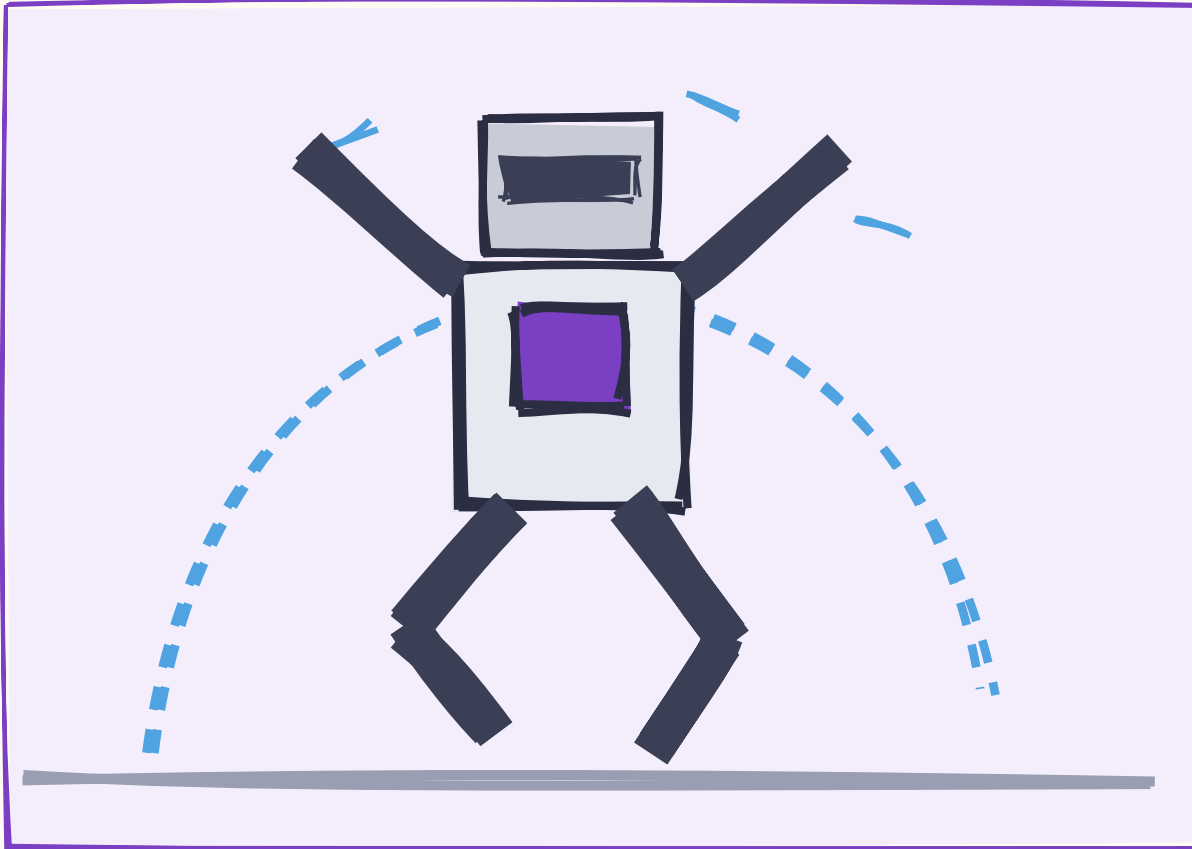
## Fascinating fact

Spot can carry about 14 kg on its back and walk on rough ground where wheels would get stuck.



Humanoid robot · Boston Dynamics, USA

# Atlas



## What it does

A robot shaped like a **human** that can run, jump and even do **backflips**. Engineers use it to learn how robots can work in places built for people.

**SENSE:** cameras and depth sensors map the ground in front of it

**THINK:** a computer plans every jump and landing

**ACT:** powerful motors in its joints move like our muscles

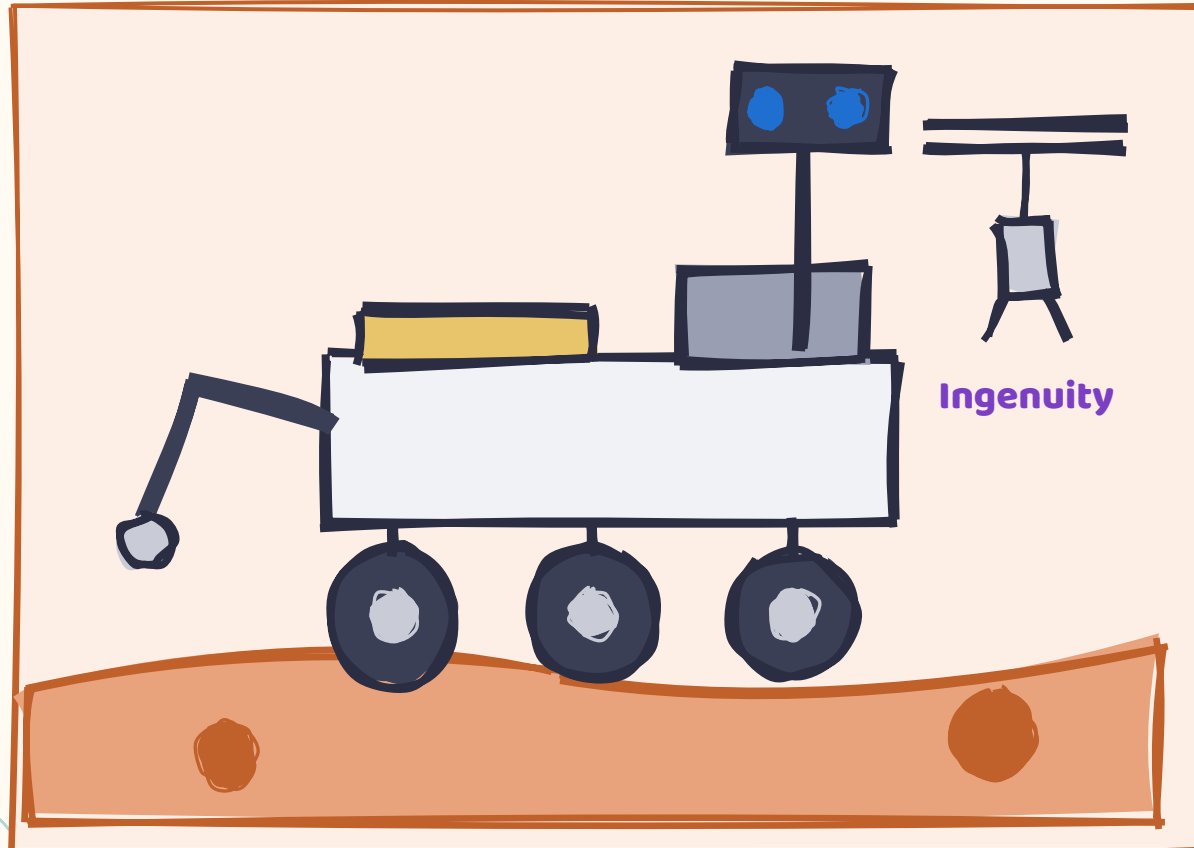
## Fascinating fact

Atlas videos doing parkour have been watched by millions of people. Balancing on two legs is one of the hardest problems in robotics!



Mars rover · NASA, USA · landed 2021

# Perseverance



## What it does

A car-sized robot **exploring Mars**. It drills rocks, collects samples and searches for signs of ancient life. It also carried **Ingenuity**, a small helicopter that flew 72 times on Mars.



**SENSE:** about two dozen cameras, a laser and science instruments

**THINK:** drives itself between targets, because radio messages take minutes to reach Earth

**ACT:** 6 wheels, a robotic arm with a drill, and sample tubes

## Fascinating fact

Perseverance weighs about 1,000 kg and is powered by a nuclear battery, not solar panels.



Inside every robot

# The 6 parts of a robot



**Sensors** = the eyes, ears and skin  
Collect information



**Controller** = the brain  
Runs the program, makes decisions



**Actuators** = the muscles  
Move, light up, make sound



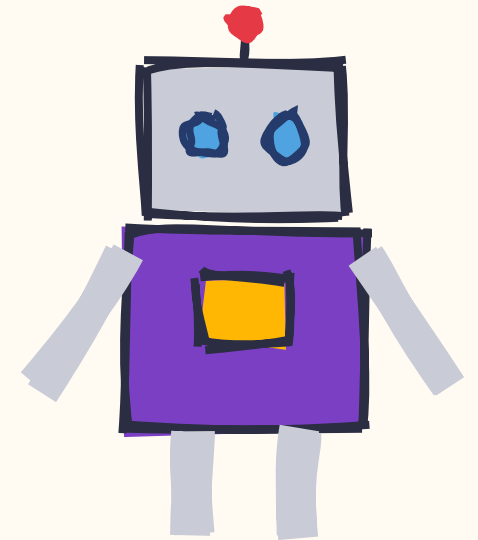
**Power** = the food and energy  
Batteries or a power supply



**Body** = the skeleton  
Holds everything together



**Program** = the thoughts  
Instructions written by us



Human vs robot

We have the same 6 parts! Our nerves are the wires.



## Sensing



# Sensors: how robots notice the world

**Ultrasonic**

Distance, using sound echoes

**Light (LDR)**

How bright or dark it is

**Camera**

Pictures, faces, colours

**Touch / button**

Being pressed or bumped

**Temperature**

How hot or cold it is

**Motion (PIR)**

A person moving nearby



Our Art Bot today has **no sensors** at all. Remember this: at the end we will ask if that means it is **not really a robot**.

**Your sensors**

Eyes, ears, nose, tongue, skin. Which one is like a camera?





Acting

# Actuators: how robots move and react

**DC motor**

Spins wheels and fans

**Servo motor**

Turns to an exact angle

**LED**

Lights up to show a signal

**Buzzer**

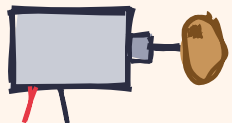
Beeps an alarm or a tune

**Screen**

Shows words and numbers

**Pump**

Moves water



Today our Art Bot uses one actuator: a **DC motor**. It turns electricity into **spinning**, and we will turn that spinning into **shaking**!

**Think**

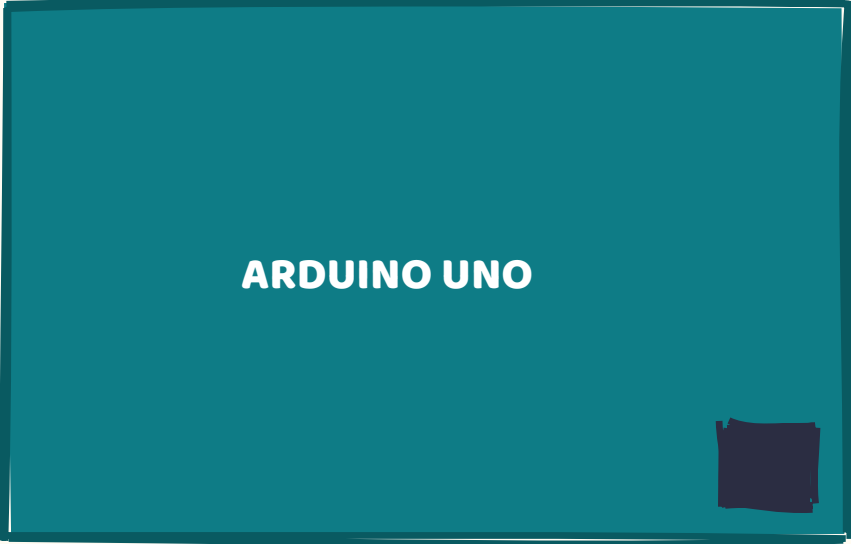
Which actuators does a robot vacuum need?





Thinking

# The brain: a microcontroller



ARDUINO UNO

## Bigger brain

Later we use the ESP32: a brain with Wi-Fi and Bluetooth built in!

### What is it?

A **microcontroller** is a tiny computer on a board. The **Arduino Uno** reads sensors, runs our program and controls actuators. It is cheap, safe (5 volts) and used by millions of learners.



### How do we program it?

1. Write the program on a laptop.
2. **Upload** it through a USB cable.
3. The Arduino runs it **by itself**, even without the laptop.



## Robots + AI

# When robots learn



## A normal robot

Follows **rules we write**.

*IF something is closer than 20 cm, THEN beep.*

## A robot with AI

Learns from **examples**.

Show it 1,000 pictures of cats and it learns to *recognise* a cat by itself.



Face unlock on phones



Cars that read road signs



Farm robots that pick ripe fruit



Voice assistants that understand you





Stand up and play!

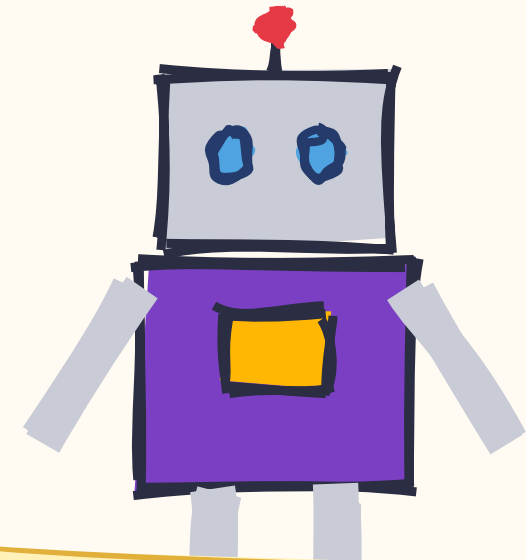
# Program a human robot

## How to play (pairs)

1. One partner is the **robot**, one is the **programmer**.
2. The robot must reach a target (a chair, a door) and pick up an object.
3. The robot follows instructions **exactly**, nothing more: "step forward", "turn right", "stop".
4. If the program is wrong, the robot does the wrong thing!



Was your program right the first time? Fixing mistakes in a program is called **debugging**.



### Robot rule

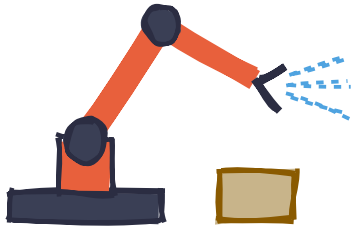
Robots do what we TELL them, not what we MEAN!





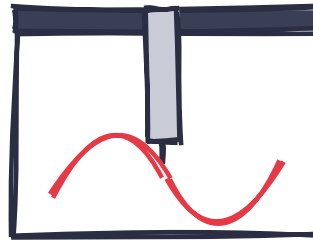
Robots + art

# Machines that make art



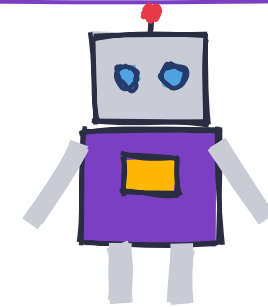
## Painting arms

Spray-paint every new car in the factory, smooth and even



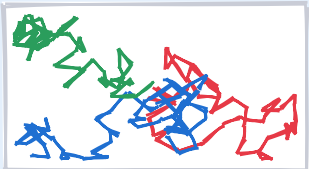
## Drawing plotters

Pens on motors draw maps, posters and designs



## Robot artists

Some robots use cameras and AI to draw portraits



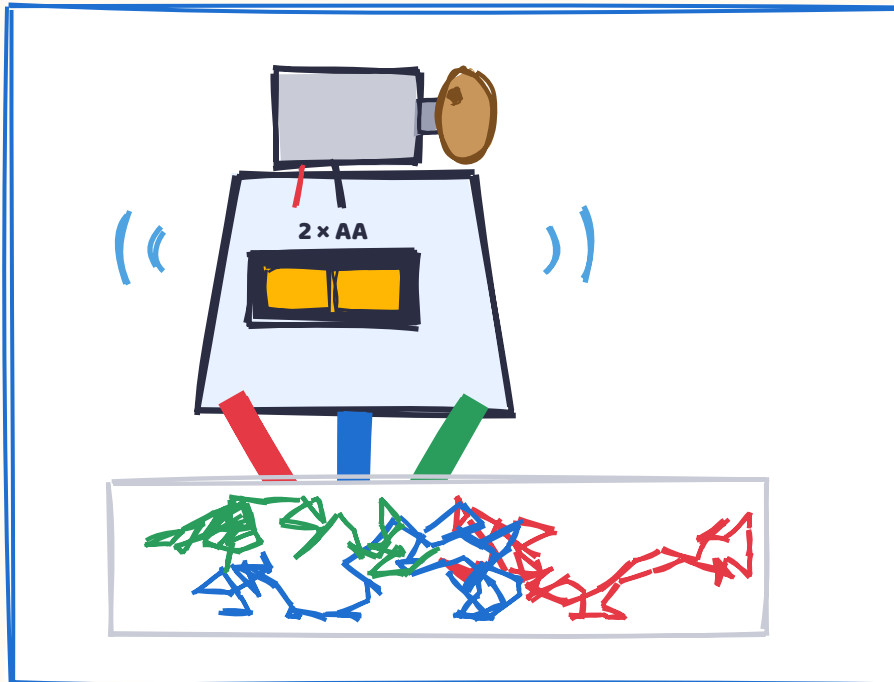
Today you build the **simplest art machine of all**: it has no brain and no plan, but its **wobble** makes a different drawing every time.





Your build

# Make an Art Bot!



## How it works

A **cup** stands upside down on **marker legs**. A **DC motor** on top spins a **cork stuck on off-centre**. The heavy side swings round and round and makes the whole bot **shake**. The shaking makes it walk, and the markers **draw its path**.



## Sense, think, act?

**Sense:** ... nothing yet!

**Think:** ... no brain yet!

**Act:** the motor shakes it and the markers draw

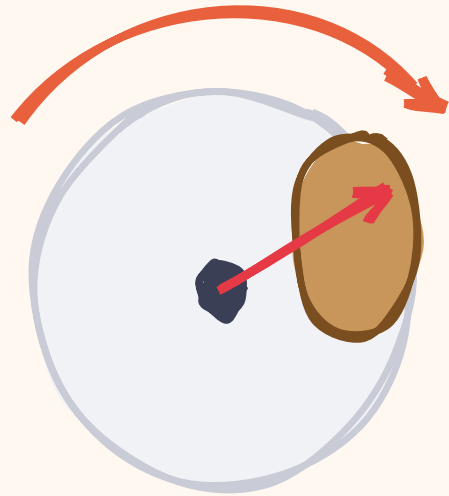
Keep this in mind  
Is our Art Bot really a  
robot? We will decide at the  
end.





The secret

# Why does it shake?



heavy side pulls it round

**Try it**  
Hold a running motor gently. Feel the buzz in your fingers!

## Off-centre weight

If the cork sits **in the middle** of the shaft, it spins smoothly: no shake. If it sits **to one side**, the heavy side pulls the motor one way, then the other way, many times every second. That pull is a **vibration**.



Your **phone** has the same thing inside! A tiny motor with an off-centre weight makes it **vibrate** when someone calls.





Build it

# Our builder's kit (per pair)

**Plastic or paper cup**

The body

**3 or 4 markers**

The legs that draw

**Small DC motor**

Makes the shake

**Cork (or eraser)**

The off-centre weight

**2 × AA + holder**

The power

**Tape + switch**

Hold it all together

**Safety rules**

1. Use a **plastic or paper cup**, never a real glass: a shaking bot can fall off the table.
2. **Only the teacher uses the hot glue gun.**
3. Keep fingers and hair away from the spinning cork.
4. Switch off or unclip the battery whenever you change something.

**Washable!**

Use washable markers and cover the table with big paper.

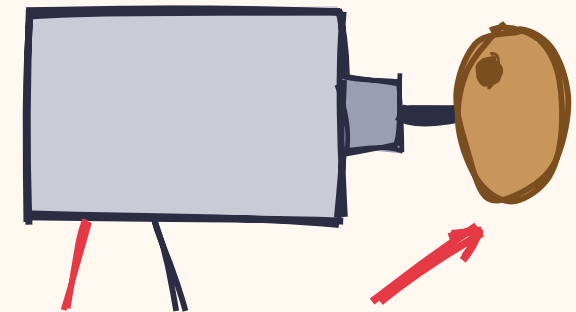


## Build step 1 of 3

# Make the shaker (motor + cork)

- a** Push the motor shaft into the cork **near its edge**, not in the middle.  
(Teacher can make a small hole first.)
- b** Check the cork is **tight**: it must not slip on the shaft.
- c** Connect the 2 motor wires to the battery holder. The cork should spin and the motor should **buzz** in your hand.
- d** Disconnect again before the next step.

## DC motor



**off-centre!**

Off-centre = shake

If the cork is in the middle, your bot will barely move.  
Move it to the edge!

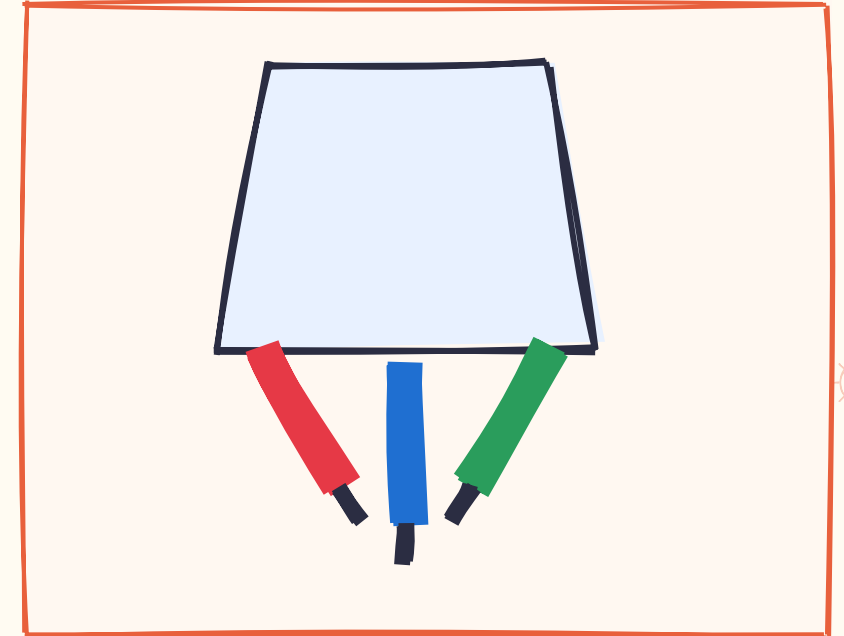


Build step 2 of 3

# Make the body (cup + marker legs)



- a** Turn the cup **upside down**.
- b** Tape 3 or 4 markers around the rim like **legs**, tips pointing down. Spread them out evenly.
- c** Take the **caps off**. Make sure all tips touch the paper when the cup stands up.
- d** The cup should stand by itself. If it falls, adjust the leg angles.



## Engineer's eye

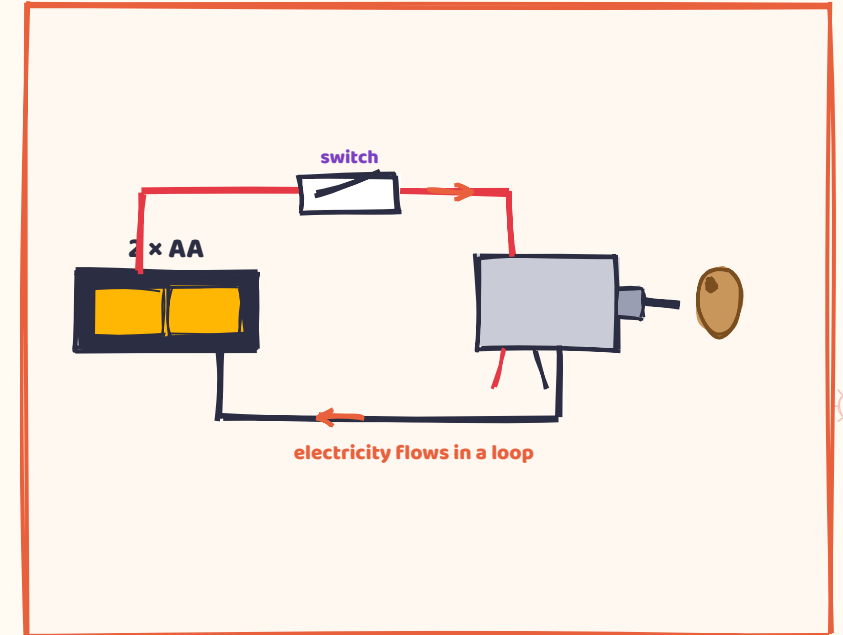
Legs spread wide = steady bot. Legs close together = wild, wobbly bot. Which do you want?



## Build step 3 of 3

# Put it together and switch on

- a** Teacher hot-glues (or you tape) the **motor on top** of the cup, with the cork hanging over the edge so it can spin freely.
- b** Tape the **battery holder** on the cup's side, balanced on the other side from the motor.
- c** Add a **switch** in one wire, or just clip the battery on and off.
- d** Place it on the big paper and switch on. **Watch it draw!**

**Not moving?**

Is the circuit a complete loop? Is the cork touching the cup? Are the batteries the right way round?



Test like an engineer

# Experiment: change ONE thing



| What we changed                   | How it moved        | What it drew              |
|-----------------------------------|---------------------|---------------------------|
| Move the cork nearer the middle   | fast / slow / spins | circles / lines / zig-zag |
| Move the cork to the very edge    | fast / slow / spins | circles / lines / zig-zag |
| Legs spread wide vs close         | fast / slow / spins | circles / lines / zig-zag |
| Add a coin on one side of the cup | fast / slow / spins | circles / lines / zig-zag |
| Use 3 legs vs 4 legs              | fast / slow / spins | circles / lines / zig-zag |

## Fair test

Change **only one thing** each time. Keep the same batteries, paper and markers. Then you know what caused the change.



## Name your style

Is your bot a Scribbler, a Spinner or a Zig-Zagger?



Can you make your Art Bot draw **big circles**? Can you make it go in a **straight-ish line**? What did you change?





Big question

# So... is our Art Bot a robot?



## Think about it

**SENSE?** No sensors. It cannot notice the edge of the paper.

**THINK?** No brain. It cannot decide anything.

**ACT?** Yes! A motor moves it.

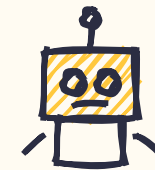
## Our answer

Not yet! It is a clever **machine**: it only ACTS. To become a robot it needs to **sense** and **think** too.



## How could we make it a robot?

Add a **light sensor** so it stops in the dark · add a **tilt or bump sensor** so it turns off when it reaches the edge of the paper · add a **tiny brain** (micro:bit or Arduino) to make it shake in patterns.





Level up!

# Challenges and art gallery



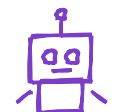
## Level 1

Make your bot draw for 30 seconds without falling over.



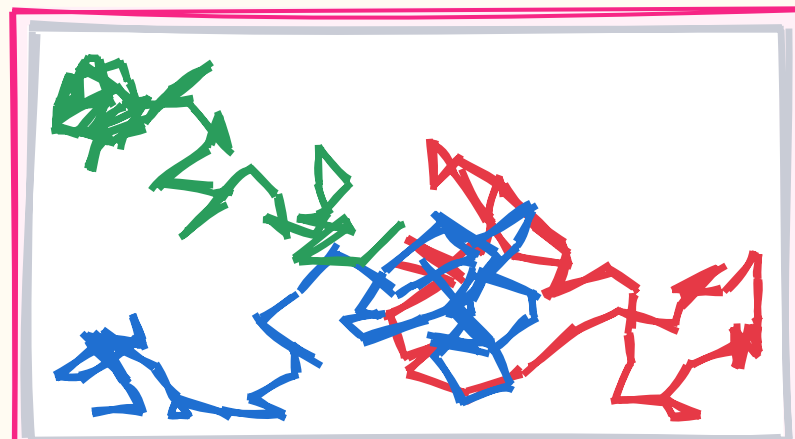
## Level 2

Design a bot that draws mostly circles, then one that draws lines.



## Level 3

Decorate it as a character: give it eyes, a name and a story!



**Gallery walk:** put every drawing on the wall with the bot that made it. Can others guess which bot drew which picture?





Coming up in the club

# What will we build next?



## Robot cars

that avoid walls and follow lines



## Robot arms

that copy your hand



## Smart devices

you control from your phone



## Smart farms

that water plants by themselves



## Safety systems

for fire, gas and security



## AI projects

that see and recognise things



Next time we add a **brain** and **sensors**, and our machines become real robots!





Show what you know

# What did we learn?



A robot senses, thinks and acts by itself



Robots have sensors, a controller, actuators, power, a body and a program



A DC motor is an actuator: it turns electricity into spinning



An off-centre weight turns spinning into shaking (vibration)



Our Art Bot only acts, so it is a machine, not yet a robot



**Art Bot Builder**

badge unlocked!



**At home**

Find 3 things at home that vibrate. Which ones have a spinning motor inside?



**Exit ticket:** what would you add to turn your Art Bot into a real robot?

